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System and method for finetuning automated sentiment analysis

Abstract

A method and system for finetuning automated sentiment classification by at least one processor may include: receiving a first machine learning (ML) model M.sub.0, pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset; associating one or more instances of model M.sub.0 to one or more corresponding sites; and for one or more (e.g., each) ML model M.sub.0 instance and/or site: receiving at least one utterance via the corresponding site; obtaining at least one data element of annotated feedback, corresponding to the at least one utterance; retraining the ML model M.sub.0, to produce a second ML model M.sub.1, based on a second annotated training dataset, wherein the second annotated training dataset may include the first annotated training dataset and the at least one annotated feedback data element; and using the second ML model M.sub.1, to classify utterances according to one or more sentiment classes.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to the field of automated natural language processing. More specifically, the present invention relates to finetuning automated sentiment classification.

BACKGROUND OF THE INVENTION

(2) Sentiment analysis is the task of detecting sentiments in written text or verbal utterances, including for example, sentiments such as happiness, anger, surprise, and sarcasm. Currently available methods for performing sentiment analysis may include, for example, training a machine learning (ML) model to differentiate sentimental sentences from neutral sentences. For an input

sentence, such a ML model may return a sentiment class and a score which may reflect its confidence level in the prediction.

(3) One commercial aspect of employing sentiment analysis may be found in the domain of contact centers. In this domain, sentiment analysis is mainly used for detecting sentiments that are expressed by customers during their interactions with a company's agent. Such interactions can take the form of chats, emails, social media, voice calls and the like. In the latter case, the term sentiment analysis may refer to analysis of text that is generated from the voice call by an automated speech recognition engine.

(4) Naturally, any sentiment classification model may produce incorrect predictions, which can be either total and objective, or subjective per the perception of the call center analyst. Inherent difficulties in deployment of sentiment analysis models or systems may arise when the system encounters a new domain, which was not included in the training material, and meets new vocabulary or words that may have a specific meaning in the new domain.

SUMMARY OF THE INVENTION

(5) Embodiments of the invention may address this challenge, by implementing an interactive process, by which users may be allowed to provide feedback on predictions made by the application, and thus affect its future predictions. Embodiments of the invention may leverage this feedback in a plurality of ways.

(6) For example, utterances with corrected output sentiment may be used to retrain or finetune the model, with the aim of generalizing the model to similar utterances in the relevant domains and businesses. The terms “retrain” and “finetune” may be used herein interchangeably, in the sense of improving a training of the model, as elaborated herein.

(7) In another example, by analyzing a random sample of predictions versus truth values, embodiments of the invention may learn to optimally tune prediction thresholds for sentiment classification for each instantiation or installation of the present invention.

(8) Embodiments of the invention may include a method of finetuning automated sentiment classification by at least one processor. Embodiments of the method may include receiving a first machine learning (ML) model M.sub.0, pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset; and associating one or more instances of model M.sub.0 to one or more corresponding sites. For each ML model M.sub.0 instance, embodiments may further include receiving at least one utterance via the corresponding site; obtaining at least one data element of annotated feedback, corresponding to the at least one utterance; retraining the ML model M.sub.0, to produce a second ML model M.sub.1, based on a second annotated training dataset, wherein the second annotated training dataset may include the first annotated training dataset and the at least one annotated feedback data element; and using the second ML model M.sub.1, to classify utterances according to one or more sentiment classes.

(9) According to some embodiments of the invention, the at least one processor may obtain the at least one data element of annotated feedback interactively, via a user interface (UI) associated with the corresponding site. The annotated feedback may include, for example, classification of the at least one utterance according to zero or more sentiment classes.

(10) Embodiments of the invention may include receiving an annotated validation dataset, that may include a plurality of utterances, and computing a corresponding plurality of confidence scores. For example, each confidence score may represent a certainty level of classification of an utterance to a sentiment class. Embodiments of the invention may sort the utterances of the validation dataset in an ordered list, according to the computed confidence scores;

(11) Embodiments of the invention may include computing, for one or more entries of the ordered list, a value of classification precision; identifying an index of the ordered list that corresponds to a required classification precision value; and determining the confidence score of the identified index as a prediction threshold for classifying utterances as belonging to the sentiment class.

(12) Embodiments of the invention may include a method of finetuning automated sentiment

classification by at least one processor. Embodiments of the method may include receiving a ML model M.sub.0, pretrained to perform automated sentiment classification of utterances, wherein model M.sub.0 may be associated with at least one prediction threshold; and associating one or more instances of model M.sub.0 to one or more corresponding sites.

(13) For at least one (e.g., each) ML model M.sub.0 instance and/or site, embodiments of the invention may receive at least one utterance via the corresponding site; obtain at least one data element of annotated feedback, corresponding to the at least one utterance; automatically calibrate the at least one prediction threshold according to a required level of precision, and based on the at least one annotated feedback data element; and use the at least one prediction threshold to classify utterances as belonging to one or more sentiment classes.

(14) According to some embodiments of the invention, obtaining the at least one data element of annotated feedback may include: presenting one or more utterances to a user, via a UI; and obtaining the annotated feedback data element from the user via the UI, wherein the annotated feedback may include classification of the presented one or more utterances according to zero or more sentiment classes.

(15) According to some embodiments of the invention, presenting one or more utterances may include: calculating a plurality of confidence scores to a corresponding plurality of utterances, according to each utterance's pertinence to a sentiment class; selecting a subset of utterances of the plurality of utterances for feedback based on their respective confidence scores; and presenting the selected subset of utterances via the UI.

(16) According to some embodiments of the invention, selecting one or more utterances for feedback may include, for each sentiment class: selecting a first subset of utterances that surpass a selection threshold from the plurality of utterances; and randomly selecting a second subset of utterances from the first subset; and presenting the second subset of utterances via the UI.

(17) According to some embodiments of the invention the prediction threshold may be specific for each sentiment class.

(18) According to some embodiments of the invention the size of the second subset of utterances may be limited by a predefined subset size, that may be specific for each sentiment class. In some embodiments the second subset of utterances may include, or may be limited to 300 utterances or less.

(19) According to some embodiments of the invention, calibrating the at least one prediction threshold may include: calculating one or more confidence scores, corresponding to the one or more utterances of the second subset, according to each utterance's pertinence to a sentiment class; sorting the plurality of utterances of the second subset in a descending ordered list, according to the confidence scores; for one or more entries of the ordered list, computing a value of classification precision based on the annotated feedback; identifying an index of the ordered list that corresponds to a required classification precision value; and determining the confidence score of the identified index as the value of the prediction threshold.

(20) In some embodiments, the predefined subset size may be selected so as to guarantee that the prediction threshold corresponds to ground truth precision having a predefined confidence value (e.g., at least 95%) within a predefined margin of error (e.g., 0.05).

(21) According to some embodiments, the sentiment classes may include, for example a positive sentiment class, a negative sentiment class and a neutral sentiment class.

(22) Embodiments of the invention may include a system for finetuning automated sentiment analysis. Embodiments of the system may be associated with a specific site and may include: (a) a non-transitory memory device storing instruction code, and (b) at least one processor associated with the memory device.

(23) According to some embodiments, the at least one processor may be configured to execute the instruction code so as to: receive a first ML model M.sub.0, pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset; receive at least

one utterance via the site; obtain at least one data element of annotated feedback, corresponding to the at least one utterance; retrain the ML model M.sub.0, to produce a second ML model M.sub.1, based on a second annotated training dataset, wherein the second annotated training dataset may include the first annotated training dataset and the at least one annotated feedback data element; and use the second ML model M.sub.1 to classify utterances according to one or more sentiment classes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter regarded as the invention is particularly pointed out and distinctly claimed in the concluding portion of the specification. The invention, however, both as to organization and method of operation, together with objects, features, and advantages thereof, may best be understood by reference to the following detailed description when read with the accompanying drawings in which:

(2) FIG. **1** is a block diagram, depicting a computing device which may be included in a system for finetuning automated sentiment classification, according to some embodiments;

(3) FIG. **2** is a block diagram, depicting a system for finetuning automated sentiment classification, according to some embodiments;

(4) FIG. **3** is a table, depicting an example of data that may be used to determine a prediction threshold, according to some embodiments;

(5) FIG. **4** is a flow diagram, depicting a method of finetuning automated sentiment classification by at least one processor, according to some embodiments; and

(6) FIG. **5** is a flow diagram, depicting another method of finetuning automated sentiment classification by at least one processor, according to some embodiments.

(7) It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

(8) One skilled in the art will realize the invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting of the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are therefore intended to be embraced therein.

(9) In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the invention. However, it will be understood by those skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the present invention. Some features or elements described with respect to one embodiment may be combined with features or elements described with respect to other embodiments. For the sake of clarity, discussion of same or similar features or elements may not be repeated.

(10) Although embodiments of the invention are not limited in this regard, discussions utilizing terms such as, for example, “processing,” “computing,” “calculating,” “determining,” “establishing”, “analyzing”, “checking”, or the like, may refer to operation(s) and/or process(es) of a computer, a computing platform, a computing system, or other electronic computing device, that

manipulates and/or transforms data represented as physical (e.g., electronic) quantities within the computer's registers and/or memories into other data similarly represented as physical quantities within the computer's registers and/or memories or other information non-transitory storage medium that may store instructions to perform operations and/or processes.

(11) Although embodiments of the invention are not limited in this regard, the terms “plurality” and “a plurality” as used herein may include, for example, “multiple” or “two or more”. The terms “plurality” or “a plurality” may be used throughout the specification to describe two or more components, devices, elements, units, parameters, or the like. The term “set” when used herein may include one or more items.

(12) Unless explicitly stated, the method embodiments described herein are not constrained to a particular order or sequence. Additionally, some of the described method embodiments or elements thereof can occur or be performed simultaneously, at the same point in time, or concurrently.

(13) Reference is now made to FIG. **1**, which is a block diagram depicting a computing device, which may be included within an embodiment of a system for finetuning automated sentiment classification, according to some embodiments.

(14) Computing device **1** may include a processor or controller **2** that may be, for example, a central processing unit (CPU) processor, a chip or any suitable computing or computational device, an operating system **3**, a memory **4**, executable code **5**, a storage system **6**, input devices **7** and output devices **8**. Processor **2** (or one or more controllers or processors, possibly across multiple units or devices) may be configured to carry out methods described herein, and/or to execute or act as the various modules, units, etc. More than one computing device **1** may be included in, and one or more computing devices **1** may act as the components of, a system according to embodiments of the invention.

(15) Operating system **3** may be or may include any code segment (e.g., one similar to executable code **5** described herein) designed and/or configured to perform tasks involving coordination, scheduling, arbitration, supervising, controlling or otherwise managing operation of computing device **1**, for example, scheduling execution of software programs or tasks or enabling software programs or other modules or units to communicate. Operating system **3** may be a commercial operating system. It will be noted that an operating system **3** may be an optional component, e.g., in some embodiments, a system may include a computing device that does not require or include an operating system **3**.

(16) Memory **4** may be or may include, for example, a Random-Access Memory (RAM), a read only memory (ROM), a Dynamic RAM (DRAM), a Synchronous DRAM (SD-RAM), a double data rate (DDR) memory chip, a Flash memory, a volatile memory, a non-volatile memory, a cache memory, a buffer, a short-term memory unit, a long-term memory unit, or other suitable memory units or storage units. Memory **4** may be or may include a plurality of possibly different memory units. Memory **4** may be a computer or processor non-transitory readable medium, or a computer non-transitory storage medium, e.g., a RAM. In one embodiment, a non-transitory storage medium such as memory **4**, a hard disk drive, another storage device, etc. may store instructions or code which when executed by a processor may cause the processor to carry out methods as described herein.

(17) Executable code **5** may be any executable code, e.g., an application, a program, a process, task, or script. Executable code **5** may be executed by processor or controller **2** possibly under control of operating system **3**. For example, executable code **5** may be an application that may finetune automated sentiment classification as further described herein. Although, for the sake of clarity, a single item of executable code **5** is shown in FIG. **1**, a system according to some embodiments of the invention may include a plurality of executable code segments similar to executable code **5** that may be loaded into memory **4** and cause processor **2** to carry out methods described herein.

(18) Storage system **6** may be or may include, for example, a flash memory as known in the art, a

memory that is internal to, or embedded in, a micro controller or chip as known in the art, a hard disk drive, a CD-Recordable (CD-R) drive, a Blu-ray disk (BD), a universal serial bus (USB) device or other suitable removable and/or fixed storage unit. Data pertaining to verbal and/or textual utterance may be stored in storage system **6** and may be loaded from storage system **6** into memory **4** where it may be processed by processor or controller **2**. In some embodiments, some of the components shown in FIG. **1** may be omitted. For example, memory **4** may be a non-volatile memory having the storage capacity of storage system **6**. Accordingly, although shown as a separate component, storage system **6** may be embedded or included in memory **4**.
(19) Input devices **7** may be or may include any suitable input devices, components, or systems, e.g., a detachable keyboard or keypad, a mouse, and the like. Output devices **8** may include one or more (possibly detachable) displays or monitors, speakers and/or any other suitable output devices. Any applicable input/output (I/O) devices may be connected to Computing device **1** as shown by blocks **7** and **8**. For example, a wired or wireless network interface card (NIC), a universal serial bus (USB) device or external hard drive may be included in input devices **7** and/or output devices **8**. It will be recognized that any suitable number of input devices **7** and output device **8** may be operatively connected to Computing device **1** as shown by blocks **7** and **8**.
(20) A system according to some embodiments of the invention may include components such as, but not limited to, a plurality of central processing units (CPU) or any other suitable multi-purpose or specific processors or controllers (e.g., similar to element **2**), a plurality of input units, a plurality of output units, a plurality of memory units, and a plurality of storage units.
(21) A neural network (NN) or artificial neural network (ANN), such as a neural network implementing a machine learning (ML) or artificial intelligence (AI) function may refer to an information processing paradigm that may include nodes, referred to as neurons, organized into layers, with links between the neurons. The links may transfer signals between neurons and may be associated with weights. A NN may be configured or trained for a specific task, e.g., pattern recognition or classification. Training a NN for the specific task may involve adjusting these weights based on examples. Each neuron of an intermediate or last layer may receive an input signal, e.g., a weighted sum of output signals from other neurons, and may process the input signal using a linear or nonlinear function (e.g., an activation function). The results of the input and intermediate layers may be transferred to other neurons and the results of the output layer may be provided as the output of the NN. Typically, the neurons and links within a NN are represented by mathematical constructs, such as activation functions and matrices of data elements and weights. A processor, e.g., CPUs or graphics processing units (GPUs), or a dedicated hardware device may perform the relevant calculations.
(22) Reference is now made to FIG. **2**, which depicts a system **100** for finetuning automated sentiment classification, according to some embodiments.
(23) According to some embodiments of the invention, system **100** may be implemented as a software module, a hardware module, or any combination thereof. For example, system **100** may be, or may include a computing device such as element **1** of FIG. **1**, and may be adapted to execute one or more modules of executable code (e.g., element **5** of FIG. **1**) to finetune automated sentiment classification, as further described herein.
(24) According to some embodiments, and as shown in FIG. **2**, system **100** may include or may receive a machine-learning (ML) based model **120** (e.g., **120**A, **120**B). ML model **120** may be initially trained, or pretrained, in an initial setting or environment **20**, to perform automated sentiment classification of utterances, using a first annotated dataset. The term "utterance" may be used in this context to refer to a verbal, or vocal utterance that may include one or more syllables or words. Additionally, or alternatively, the term "utterance" may be used in this context to refer to a textual data element that may represent one or more uttered syllables or words, and may have been obtained via a speech-to-text conversion, as known in the art.
(25) As depicted in FIG. **2**, the initial, or pretrained model may be denoted herein as a "base

model", M.sub.0, and/or ML model **120**, and these notations may be used herein interchangeably.
(26) The first annotated dataset may be, or may include for example an annotated or labeled training dataset, denoted as element **23** in FIG. **2**. Additionally, or alternatively, the first annotated dataset may be, or may include a validation dataset denoted as element **25** in FIG. **2**.
(27) According to some embodiments, initial setting, or environment **20** may refer herein to an environment that may not necessarily be associated with a specific client environment. For example, initial setting, or environment **20** may pertain to a vendor or distributor of system **100**, and may be disconnected, or offline, in relation to any specific client environment. In a complementary manner, the term client site or instantiation environment **30** may be used herein to refer to a site or a computing environment that may pertain to a specific installation or instantiation of system **100** on one or more computing devices (e.g., computing device **10** of FIG. **1**), that are related to a client site.
(28) As illustrated by the thick arrow of FIG. **2**, after the initial training of system **100** in the initial setting or environment **20**, system **100** may be installed, or deployed on one or more computing devices **10** that may be related to a client site, including for example an organizational computing network, one or more computers hosting a website associated with a specific client or organization, and the like. It may thus be said that one or more installations or instantiations of system **100**, including one or more respective instances of model M.sub.0 may be associated to one or more corresponding instantiation environments or client sites **30** (e.g., websites, computing networks, etc.).
(29) According to some embodiments, for one or more (e.g., each) instance of ML model M.sub.0 (e.g., pertaining to each client site **30**), system **100** may receive at least one utterance **143** via the corresponding client site **30**.
(30) For example, client site **30** may be or may include a chat room of a specific client or organization, a website associated with a client or organization, a contact center of the client or organization, and the like. Accordingly, received utterance **143** may be a vocal utterance, a verbal utterance, a textual element representing a verbal utterance, etc. that has been received (e.g., via phone, chat, etc.) at client site **30**.
(31) In one non-limiting example, system **100** may be installed or deployed in a client site **30** as part of a client site as a Contact Center Analytics application. As elaborated herein, system **100** may be adapted to enable users of the Contact Center Analytics application to affect the process of sentiment content detection by providing interactive feedback to the application on its predictions through a user interface **140**. The interactive feedback may facilitate a local, independent finetuning or retraining of ML model **120**, in a sense that it may be associated with a specific client site **30** (e.g., a specific contact center site or Cloud tenant). For example, the finetuning or retraining of ML model **120** may adapt the sentiment analysis model **120** to a jargon that may be used in an organization's domain and/or business. According to some embodiments, users may be able to fix unsatisfactory predictions of ML model **120**, and/or strengthen correct predictions.
(32) As elaborated herein, system **100** may use interactive feedback to classification of utterances to: (a) retrain or finetune ML model **120**, and (b) finetune one or more prediction thresholds, to yield a desired target precision.
(33) According to some embodiments, system **100** may include a user interface (UI) **140**, that may correspond to, or facilitate an interface to client site **30**. For example, UI **140** may be an application (e.g., a webpage) that may facilitate an interface to a client website **30**. Additionally, or Alternatively, UI **140** may be or may include a chat application, that may facilitate a client site chat room **30**. Additionally, or Alternatively, UI **140** may be or may include an application that may facilitate a call center, between representatives of an organization and customers of that organization.
(34) According to some embodiments, system **100** may receive, e.g., via UI **140**, one or data elements that include unseen data such as unseen utterances **143**. System **100** may perform one or

more on-site predictions on the unseen utterances data elements **143**. In other words, system **100** may utilize instantiation **120**B of pretrained model M.sub.0 to classify or predict (as commonly referred to in the art) a class at least one received utterance **143**.

(35) The predicted classes may be, or may include for example at least one first class of positive utterances, such as utterances in which a customer expresses content with a service provided by the client site (e.g., "thanks a lot, I appreciate your helpfulness"). In a complementary manner, the predicted classes may include at least one a second class of negative utterances such as utterances in which a customer expresses discontent with a service provided by the client site (e.g., "I have been waiting for a very long time").

(36) According to some embodiments, system **100** may receive or obtain, e.g., from a user via UI **140**, at least one feedback data element **145**. Feedback data element **145** may be an annotated feedback data element, corresponding to the at least one utterance **143** and/or to a corresponding on-site prediction. In other words, the at least one data element of annotated feedback may be obtained interactively, via a user interface (UI) associated with the corresponding client site **30**, as elaborated herein.

(37) For example, UI **140** may present to a user (e.g., via output device **8** of FIG. **1**, such as a screen) at least one utterance **143** and a corresponding on-site prediction **320** made by model **120**. The user may disagree with on-site prediction **320**. For example, the user may determine that system **100** has erroneously classified the utterance **143** as a negative sentiment, where in fact it was positive or neutral, and vice versa. The user may subsequently choose to input (e.g., via input device **7** of FIG. **1**) an interactive feedback data element **145**, to correct this error. For example, the user may include, in the annotated feedback, classification of the at least one utterance according to zero, one or more sentiment classes, such as a "positive sentiment" class **121**, a "negative sentiment" class **121** and a neutral class **121**.

(38) The term "interactive" may be used herein in a sense that system **100** may present, and/or query the validity of on-site prediction **320**, in relation to utterance **143** via UI **140**, and may subsequently receive feedback data element **145** from a user, via UI **140**, in response to the presentation or query.

(39) In some embodiment, e.g., where the sentiment classes **121** include a "positive sentiment" class and a "negative sentiment" class, feedback **145** may include, for each class **121** (e.g., "positive sentiment" and "negative sentiment"), one or more feedback types, including for example a false positive (FP) feedback type, a false negative (FN) feedback type, a true positive (TP) feedback type, and a true negative (TN) feedback type.

(40) For example, a false-positive (FP) feedback type may relate to either a negative or positive sentiment class, and may be denoted herein as FP/NEG and FP/POS, respectively. For example, if an utterance **145** such as "the weather today is awful" was erroneously classified **121** by model **120** as conveying a negative sentiment, a user may provide annotating interactive feedback **145** such as FP (e.g., FP/NEG). Alternatively, if an utterance such as "thanks, goodbye" was erroneously classified **121** by model **120** as conveying a positive sentiment, a user may provide annotating interactive feedback **145** such as FP (e.g., FP/POS).

(41) In another example, a false-negative (FN) feedback type may relate to either a negative or positive sentiment class, and may be denoted herein as FN/NEG and FN/POS, respectively. For example, the utterance or sentence **143** "if you don't solve my problems, I am gonna switch to Netflix" may be annotated by interactive feedback **145** as FN (e.g., FN/NEG) if it was not classified **121** by model **120** as conveying a negative sentiment. Alternatively, an utterance **143** such as the sentence "thanks, I really appreciate it" may be marked by the user as FN (FN/POS) if it was not classified **121** by model **120** as conveying a positive sentiment.

(42) In another example, true-positive (TP) annotation and true-negative (TN) annotations may be provided as interactive feedback **145** to support a prediction made by model **120**, for either one of the sentiment classes (e.g., negative, positive, or neutral).

(43) According to some embodiments, system **100** may include a training module **130**, adapted to retrain or finetune ML model **120** (e.g., M.sub.0), so as to produce a second ML model, or a second version of ML model **120**. This finetuned model is denoted in FIG. **2** as model M.sub.1, or element **120**B.
(44) According to some embodiments, training module **130** may retrain or finetune ML model **120** based on an annotated training dataset, that may include the first annotated training dataset (e.g., training dataset **23** and/or validation dataset **25**), as well as the at least one interactive, annotated feedback data element **145**.
(45) According to some embodiments, the sentiment classes may represent a sentiment, and may include for example, a positive sentiment class, a negative sentiment class, a neutral sentiment class, etc. System **100** may utilize the fine-tuned or retrained ML model M.sub.1, to classify incoming utterances **143** according to one or more sentiment classes (e.g., positive sentiment, negative sentiment, neutral, etc.). Additionally, or alternatively, system **100** may utilize training module **130** to continuously train or fine-tune ML model M.sub.1 **120**B. The term "continuously" may be used in this context in a sense that training module **130** may repeatedly or periodically retrain ML model M.sub.1 as elaborated herein, following for example ingress of one or more interactive feedback data elements **145**, and the like.
(46) It may be appreciated by a person skilled in the art that that finetuned model M.sub.1 may be a product of three factors: (a) interactive feedback **145**, (b) base model M.sub.0 (e.g., denoted **120**A) and (c) the original dataset (e.g., training dataset **23** and/or validation dataset **25**). According to some embodiments, training module **130** may retrain or finetune ML model **120** based on (a) and on one or more of (b) and (c).
(47) It may be appreciated that input for any finetuning process may include feedback data. However, a model that is finetuned based solely on feedback will tend to forget parts of the original training material. Therefore, the original training data or parts of it are often added to the finetuning training set.
(48) According to some embodiments, training module **130** may include at least part (e.g., all) of the original training dataset (e.g., training dataset **23** and/or validation dataset **25**) as input for the finetuning process. For example, training dataset **23** and/or validation dataset **25** may be stored on a cloud-based storage server, to facilitate training of local (e.g., pertaining to client site **30**) instantiations of system **100**.
(49) According to some embodiments, portions of data used for training may be weighted, so as to assign prominence to any specific portion. For example, interactive user feedback data **145** may be weighted or duplicated during the process of retraining or finetuning of model **120**, to give extra importance to user feedback **145**. Experimental results have shown that 5 to 10-fold duplication of user feedback **145** for retraining model **120** may provide optimal performance.
(50) According to some embodiments, training module **130** may commence the finetuning process of model **120** from the already trained, on-site model (e.g., M.sub.0 or M.sub.1) **120** rather than "from scratch" (e.g., using random weights). Experimental results have shown that this manner of training may produce results that are stable throughout the finetuning process, as interactive feedback data **145** is continuously input. In contrast, training of model **120** from scratch may tend to converge into different local optima in different runs, resulting in more fluctuations in scores of utterances that are not related to feedback data **145**. Training module **130** may measure this stability by the mean change of F1 in subsequent finetuning invocations, each time increasing the number of feedback **145** data instances by a few more samples. Training module **130** may use a small learning rate (e.g., in the order of 3.sup.e-5), which may, for example, be 3 to 30 times smaller than the learning rate that was used to train base model M.sub.0.
(51) Table 1 below includes examples of feedback utterances that were used to finetune a base model M.sub.0 (e.g., so as to produce finetuned model M.sub.1) as well as variants of these utterances, on which finetuned model M.sub.1 succeeded to make accurate predictions, but the base

model M.sub.0 failed. Predictions of Table 1 are considered binary, in a sense that they may either be correct or incorrect, with respect to a prediction threshold that attains precision of 80% on the validation set for the respective model (e.g., M.sub.0 and M.sub.1). In the examples of Table 1, the finetuned model M.sub.1 was able to nicely generalize classification **121** of utterances **143** (e.g., sentences) that have similar meaning to the original utterances but are not exactly the same.

(52) TABLE-US-00001 TABLE 1 Feedback type Feedback utterance Similar utterance FN/NEG kinda tried to cheat me you they actually tried to cheat know me FN/NEG we've been pulling our hair i have been pulling my hair out trying to figure this out out trying to understand this ever since FP/NEG we didn't know if it was part a power outage has been of the you know the outage going on i guess that's been going on i guess FP/NEG i feel such pain for him i feel so much pain for them

(53) An important decision in any ML-based classification process is how to predict a class (e.g., produce a class prediction) from a raw output vector of the classifier. This process may require computing a set of prediction thresholds, which is commonly referred to as a working point. The working point may be determined by finding the prediction threshold values that yield a required precision value, p_req, over a validation dataset **25**. The precision may be calculated, for each class, as the number of correct predictions divided by the number of positive predictions, as elaborated in equation Eq. 1, below:

Precision=$TP/(TP+FP)$ Eq. 1: where TP is the true-positive ratio of the relevant class, and FP is the false-positive ratio of the relevant class.

(54) For example, for each class (e.g., a negative sentiment class) a calculated precision value, p_calc, at prediction threshold 0.8 may be the number of correct predictions (e.g., predictions of negative sentiment) with confidence>=0.8, divided by the total number of predictions (e.g., correct, and incorrect predictions of negative sentiment) with confidence>=0.8.

(55) The problem with this approach may be that validation set **25** may not necessarily represent data (e.g., utterance data elements **143**) on client site **30**. Therefore, the actual, calculated precision value p_calc may be lower or higher than required precision p_req.

(56) If the actual, calculated precision p_calc of a specific class is lower than p_req (e.g., beyond a predefined margin value), then it may be desired to increase p_calc so as to reduce the number of incorrect predictions of system **100**, and avoid having too many erroneous predictions being highly ranked by system **100**, as elaborated herein.

(57) In a complementary manner, if p_calc of a specific class is higher than p_req (e.g., beyond a predefined margin value) then it may be desired to decrease the prediction threshold of the relevant class, in order to increase the recall of the class, and obtain more high-quality classifications. As known in the art a recall value of a class may be calculated according to equation Eq. 2, below:

Recall=$TP/(TP+FN)$, EQ. 2: where TP is the true-positive ratio of the relevant class, and FN is the false-negative ratio of the relevant class.

(58) According to some embodiments, ML model **120** may be configured to receive an input utterance **143**, and produce (or predict, as commonly referred to in the art) a sentiment class **121** and a corresponding confidence score **122**. Confidence score **122** may represent a level of confidence, or certainty of classification of the relevant utterance to a sentiment class. Confidence score **122** may be or may include a numerical value, e.g., between 0 and 1, where a low value (e.g., 0.1) represents low confidence of classification **121**, and a high value (e.g., 0.9) represents high confidence of classification.

(59) In some embodiment, e.g., where the sentiment classes **121** include a “positive sentiment” class and a “negative sentiment” class, the majority of predictions may be neutral, e.g., neither pertaining to a positive sentiment nor to a negative sentiment. Therefore, a practical challenge in such embodiments may be to detect polar (e.g., negative, or positive) utterances.

(60) To this end, initial prediction threshold module **150**A may set an initial working point by finding two prediction thresholds **151**A, one for the negative class and one for the positive class,

that yield a required level of precision (e.g., p_req=0.8) on validation set **25**. However, as discussed herein, this approach may not guarantee the required precision in view of unseen utterance data on client site **30**. To address this problem, system **100** may include a finetuning threshold module **150**B, adapted to compute prediction thresholds **151**B based on client site **30** specific data utterances **143** and corresponding confidence scores **122**, as elaborated herein.

(61) According to some embodiments of the invention, system **100** may exploit an insight that the calculated precision, p_calc, may only depend on utterances **143** that surpass prediction threshold **151**B. Therefore, finetuning threshold module **150**B may suffice in sampling a relatively small random subset of utterances **143** from the overall number of incoming utterances **143**, which have high polar confidence scores **122**. For this subset, system **100** may actively prompt (e.g., via UI **140**) a user for feedback. The feedback may, for example be in a form of a correct/incorrect selection, to confirm or refute a classification of a relevant utterance **143**. As elaborated herein, finetuning threshold module **150**B may then look for prediction thresholds **151**B (e.g., for negative/positive sentiment classes) that will produce the required precision p_req on this sample. As the sampling is performed in random, the actual, calculated precision value p_calc for the client site **30** utterance data **143** may be guaranteed to lie within an arbitrarily narrow confidence interval from the target precision p_req. The higher the sample size, the smaller the margin of error may be.

(62) It may be appreciated by a person skilled in the art of statistics, and has been experimentally demonstrated by embodiments of the invention, that a subset of sampled utterances **143** that may be as narrow as 250 samples (per each polar class) may ensure a 95% confidence interval with a margin of error that may be smaller than 0.05. In other words, embodiments of the invention may perform an intelligent selection of a very narrow subset of samples of utterance data **143** for interactive feedback **145**, and thus require minimal effort of human input, while maintaining a high confidence interval, with a small margin of error.

(63) As elaborated herein, finetuning threshold module **150**B may perform client site **30**-specific tuning of prediction thresholds **151**B, based on random sampling of utterances **143** with high polar scores. The computation of different prediction thresholds **151**B (e.g., negative sentiment threshold and positive sentiment threshold) may be performed similarly, and independently.

(64) According to some embodiments, finetuning threshold module **150**B may be adapted to determine a prediction threshold value TH (denoted in FIG. **2** as element **151**B), that may induce a predefined confidence interval which is likely to contain a required target precision value p_req, with a predefined margin of error. For example, finetuning threshold module **150**B may aim to find a 95% confidence interval around p_req=0.8, with a margin of error e<0.05. It may be appreciated by a person skilled in the art of statistics, and has been shown that such an interval may be guaranteed if the interactive feedback **145** includes at least 250 random samples, taken from the range of confidence scores [TH,1].

(65) However, since the exact prediction threshold TH **151**B value may not be known in advance, and as it may theoretically lie in the range [0.5,1], acquiring 250 samples in the critical range [TH,1] may require drawing many irrelevant samples from the range [0.5,TH). For example, if the confidence scores of predictions of model **120** are uniformly distributed in the range of [0.5,1], and if prediction threshold TH=0.75 yields the target precision p_req, then finetuning threshold module **150**B may require 500 samples to compute the value of prediction threshold TH **151**B, even though only 250 of them serve to estimate the precision above prediction threshold TH **151**B. As every sample may require feedback **145** from a user, via UI **140**, it may be desirable to minimize the number of samples needed beyond the minimum of 250 that are required to guarantee the confidence interval.

(66) As elaborated herein, (e.g., in relation to FIG. **2**), system **100** may include or may receive a ML model **120**, pretrained to perform automated sentiment classification of utterances **143**. ML model **120** may be associated with at least one initial prediction threshold **151** (e.g., **151**A).

(67) According to some embodiments, one or more instances of ML model **120** may be instantiated

or associated with to one or more corresponding client sites **30**. For each ML model **120** instance (and each corresponding client site **30**), system **100** may receive at least one utterance **143** via the corresponding site, obtaining at least one data element of interactive annotated feedback **145**, corresponding to the at least one utterance **143**.

(68) As elaborated herein, for each ML model **120** instance, system **100** may subsequently automatically calibrate the at least one prediction threshold **151** (e.g., **151**A) to obtain a calibrated, or fine-tuned prediction threshold **151**B, according to a required level of precision p_req, and based on the at least one annotated feedback data element.

(69) ML model **120** may then utilize the at least one fine-tuned prediction threshold **151** (e.g., **151**B) to classify utterances **143** as belonging to one or more sentiment classes **121**.

(70) For example, if ML model **120** predicts a classification **121** (e.g., “positive sentiment”) of a specific utterance **143**, and the corresponding confidence score **122** exceed the prediction threshold **151**B for that class, then system **100** may determine that the relevant utterance is indeed classified as predicted by ML model **120** (e.g., “positive sentiment”). In a complementary manner, if the corresponding confidence score **122** does not exceed the prediction threshold **151**B for that class, then system **100** may determine that the relevant utterance is not classified as predicted by ML model **120** (e.g., “neutral sentiment”).

(71) Reference is now made to FIG. **3** which is a table, depicting an example of data that may be used to determine a prediction threshold **151**B, according to some embodiments of the invention. It may be appreciated that the data depicted in FIG. **3** may represent a simplified algorithm for automatic calibration of the at least one prediction threshold **151**B. A more elaborated version of the automatic prediction threshold algorithm is also provided herein.

(72) As elaborated herein (e.g., in relation to FIG. **2**), system **100** may present to a user e.g., via UI **140** one or more utterances **143** and a corresponding on-site predictions **320** made by model **120** (e.g., model M.sub.0 or M.sub.1). For example, an utterance may be verbal or textual utterance of a negative sentence, such as: “I am not pleased with the service”, and the on-site prediction **320** made by model **120** may include an indication of a “negative sentiment” classification.

(73) According to some embodiments, model **120** may calculate a plurality of confidence scores **122**, corresponding to a plurality of utterances **143**, according to each utterance's pertinence to a sentiment class. For example, as shown in the left-most column in the example of FIG. **3**, model **120** may calculate negative (e.g., negative sentiment) class confidence scores, as a floating number in the range of [0, 1], where high values (e.g., 0.9) represent a negative sentiment classification with high confidence and low values (e.g., 0.1) represent a negative sentiment classification with low confidence.

(74) According to some embodiments, processor **110** may select a subset of utterances **143** of the plurality of utterances **143**, to be presented on UI **140** for feedback, based on their respective confidence scores **122**. UI **140** may subsequently present the selected subset of utterances **143** and obtain feedback therefrom. For example, as shown in the “Human annotation” column of the example of FIG. **3**, a human user may provide this feedback as a yes or no (denoted 0/1) annotation, indicating whether they agree (e.g., ‘1’) or disagree (e.g., ‘0’) to the classification.

(75) According to some embodiments, processor **110** may select the subset of utterances **143** for feedback by: (a) selecting a first subset of utterances which have a confidence score **122** that surpasses a selection threshold (e.g., 0.5) from the plurality of utterances; and (b) randomly selecting a second subset of utterances from the first subset.

(76) As explained herein, as the selection of utterances for feedback is performed randomly, the actual precision value for the client site **30** utterance data **143** may be guaranteed to lie within a narrow confidence interval from computed precision p_calc. As elaborated herein, embodiments of the invention may fine-tune the precision of classification prediction to converge to a required precision, denoted p_req. Therefore, the actual precision value for the client site **30** utterance data **143** may be guaranteed to lie within a narrow confidence interval from requested precision p_req.

In addition, as the sample size (e.g., the number of utterances in the second subset) increases, the smaller the margin of error may be.

(77) According to some embodiments, the selection threshold may be set to a different or specific value for each class. For example, for a first (e.g., “negative sentiment”) class a first selection threshold value may be used, whereas for a second (e.g., “positive sentiment”) class a second, different selection threshold value may be used.

(78) As shown in FIG. **3**, finetuning threshold module **150**B may sort the plurality of utterances **143** of the second subset in a descending ordered list, according to the confidence scores **122** (e.g., the “Negative class confidence score” of the leftmost column). Finetuning threshold module **150**B may then compute, for one or more (e.g., all) entries of the ordered list, a value of classification precision based on the annotated feedback.

(79) For example, starting from the top-most confidence scores **122**, finetuning threshold module **150**B may aggregate the number of “correct” (e.g., ‘1’) human annotations (e.g., as shown in the “Aggregate corrects” column), and divide them by the ad-hoc number of utterances (e.g., as shown in the “Prefix size” column). Thus, for example, the calculated precision after four utterances is 4/4=1.0, whereas the calculated precision after five utterances is ⅘=0.8.

(80) According to some embodiments, finetuning threshold module **150**B may identify an index of the ordered list that corresponds to a required classification precision value. For example, a required classification precision value may be 0.8, and finetuning threshold module **150**B may identify the first index of an utterance, for which the calculated precision has fallen beneath the required classification precision value. In the example of FIG. **3** this is the 17.sup.th utterance, for which the calculated precision has fallen to 0.76, e.g., beneath the predefined threshold of 0.8. The utterances for which the calculated precision is below the predefined threshold are marked in gray in the example of FIG. **3**.

(81) According to some embodiments, finetuning threshold module **150**B may calibrate, or finetune prediction threshold **151**B according to the identified index.

(82) For example, finetuning threshold module **150**B may determine the calibrated, or fine-tuned prediction threshold **151**B to be equal to the confidence score **122** corresponding to the identified index. In the example of FIG. **3**, this value is 0.895903877. Alternatively, finetuning threshold module **150**B may determine the fine-tuned prediction threshold **151**B to be equal to the confidence score **122** corresponding to the index just above the identified index. In the example of FIG. **3**, this value is 0.896371917. Alternatively, finetuning threshold module **150**B may determine the fine-tuned prediction threshold **151**B to be a function (e.g., an average) between two or more class confidence scores **122** (e.g., AVG(0.896371917, 0.895903877)=0.896137897). Other specific manners of determining fine-tuned prediction threshold **151**B based on the confidence scores **122** of the sorted list may also be possible.

(83) It may be appreciated that the calibration or fine-tuning of fine-tuned prediction threshold **151**B may be implemented or performed independently for each class.

(84) Additionally, or alternatively, the prediction threshold **151**B may be specific for each sentiment class.

(85) For example, a first prediction threshold value **151**B may be used to classify utterances **143** as pertaining to a “positive sentiment” class, and a second, different prediction threshold value **151**B may be used to classify utterances **143** as pertaining to a “negative sentiment” class.

(86) According to some embodiments, the size of the second subset, e.g., the number of utterances that are selected to be presented for feedback (e.g., presented for human annotation) may be selected so as to guarantee that the prediction threshold will correspond to ground truth precision having a predefined confidence value, within a predefined margin of error. For example, the size of the second subset may be selected so as to guarantee that the prediction precision will be in 95% confidence within a margin of error of 0.05 from the ground truth precision.

(87) According to some embodiments of the invention, the size of the second subset of utterances

(e.g., utterances randomly selected for feedback) may be limited by a predefined subset limit size value, which may be specific for each sentiment class. In other words, system **100** may randomly select a first number of utterances for feedback for a first (e.g., "positive sentiment") class, and randomly select a second, different number of utterances for feedback for a second (e.g., "negative sentiment") class.

(88) According to some embodiments the second subset of utterances may include 300 utterances or less. In other words, the subset limit size value for one or more (e.g., all) classes may be 300 or less. It has been experimentally demonstrated that such a small number of 300 utterances or less may be adequate to a required precision value of p_req=0.8, at a 95% confidence interval around p_req=0.8, with a margin of error e<0.05, for a commercial-size dataset of utterances.

(89) Additionally, or alternatively, embodiments of the invention may calibrate the at least one prediction threshold **151** (e.g., **151**A) to obtain a calibrated, or fine-tuned prediction threshold **151**B, according to a required level of precision p_req, based on an annotated validation dataset **25**.

(90) In such embodiments, system **100** may receive an annotated validation dataset **25** that may include a plurality of utterances. Annotated validation dataset **25** may be annotated in a sense that it may include ground-truth classification or labeling of one or more (e.g., all) utterances included therein. Model **120** (e.g., M.sub.0 or M.sub.1) may compute a plurality of confidence scores **122**, corresponding to the plurality of utterances of annotated validation dataset **25**, where each confidence score **122** represents a certainty level of classification of an utterance to a sentiment class (e.g., a "negative sentiment" class). Finetune threshold module **150**B may subsequently sort the utterances of annotated validation dataset **25** in an ordered list, according to the computed confidence scores **122**, as depicted in the example of FIG. **3**. For one or more entries of the ordered list, finetune threshold module **150**B may compute a value of classification precision, and identify an index of the ordered list that corresponds to a required classification precision value, in a similar manner as elaborated above. Finetune threshold module **150**B may subsequently determine the confidence score **122** of the identified index as a prediction threshold **151**B for classifying utterances as belonging to the relevant sentiment class, in a similar manner as that described above.

(91) Additionally, or alternatively, finetune threshold module **150**B may calibrate a prediction threshold **151**B of at least one class (e.g., a "negative sentiment class") according to an alternative algorithm, as elaborated below.

(92) Finetune threshold module **150**B may receive, as input: (a) a target or requested precision value p_req (e.g., 0.8), (b) a minimal sample size MS (e.g., 250), (c) an extra number of samples eps (e.g., 30), which may be used to verify that below the returned prediction threshold **151**B the precision indeed falls below p_req, (d) a confidence interval probability C (e.g., 95%), and (e) a margin of error e (e.g., 0.05).

(93) The term "sample" may be used herein to refer to one or more utterances which may be selected by embodiments of the invention to be presented to a user, for obtaining supervisory feedback on classification (e.g., True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN)), as elaborated herein.

(94) According to some embodiment, finetune threshold module **150**B may employ a three-phase analysis algorithm to determine, produce or emit, based on this input, a prediction threshold value **151**B for at least one class (e.g., "negative sentiment" class or "positive sentiment" class) that achieves a precision of p_req (or as close as possible to p_req) on a random sample of at least MS samples, above prediction threshold **151**B and that guarantees a C confidence interval with margin of error e around p_req for the entire population, as elaborated herein.

(95) According to some embodiments, finetune threshold module **150**B may initially collect all utterances with a confidence score **122** that exceeds a predefined selection threshold, sort them by decreasing scores (e.g., as presented in FIG. **3**) and divide them to a predefined number n_bins of bins of equal size, denoted b.sub.1 . . . ,b.sub.n_bins, where b.sub.1 is the highest scoring bin, and b.sub.n_bins (e.g., b.sub.10) is the lowest scoring bin. For example, the number of all utterances

that have a confidence score **122** that exceeds a predefined selection threshold (e.g., >0.5) may be 600,000. Finetune threshold module **150**B may divide these utterances to 10 bins (e.g., b.sub.1 . . . ,b.sub.10) of 60,000 utterances each, where b.sub.1 is the highest scoring bin, and b.sub.10 is the lowest scoring bin.
(96) In a first phase, denoted herein as “phase 1”, finetune threshold module **150**B may perform initial estimation of the bin index where the desired prediction threshold value **151**B may be located. In this phase, T may indicate a current bin index, and ‘s’ may indicate an initial sample size per bin (e.g., 30 samples per each bin).
(97) During phase 1, and starting from the first bin (e.g., T initialized to ‘1’), finetune threshold module **150**B may repeat steps 1a, 1b and 1c below, for each bin, until a stop condition is met, as elaborated below:
(98) Phase 1
(99) {
(100) 1a. Compute the precision p_calc of all samples (e.g., from all bins) taken so far.
(101) 1b. If p_calc>=p_req, then sample s (e.g., 30) utterances from the next “unused” bin, b.sub.i, and increment i (e.g., i=i+1).
(102) The stop condition may be, for example: (a) the ad-hoc calculated precision (e.g., calculated as explained in relation to the example of FIG. **3**) p_calc is smaller than the required p_req precision; and (b) the number of samples n_samp exceeds a minimal value, as elaborated in equation 3, below:

$$p\_\text{calc} < p\_req;$$
$$\text{size}(n\_\text{samp}) >= MS + eps; \qquad \text{Eq. 3}$$

(103) For example, assuming that (p_calc<p_req) occurs on the sixth bin (e.g., i=6), and that s=30, then n_samp=6*30=180<MS=250. After adding a first batch of 10 samples for each used bin: n_samp=6*(30+10)=240<MS=250, and the stop condition is not yet met. After adding a second batch of 10 samples for each used bin: n_samp=6*(30+10+10)=300>MS=250, and the stop condition is met.
(104) In a second phase, denoted herein as “phase 2”, finetune threshold module **150**B may find the exact prediction threshold **151**B value, where the calculated precision falls below p_req.
(105) In this phase, finetune threshold module **150**B may repeat steps 2a through 2d, as elaborated below:
(106) Phase 2
(107) {
(108) 2a. Finetune threshold module 150B may sort the n_samp (e.g., 300 in the current example) acquired samples in decreasing order of confidence scores 122.
(109) 2b. For every prefix of the ordered set, finetune threshold module 150B may compute its precision p_calc, as elaborated h erein (e.g., in relation to FIG. **3**).
(110) 2c. Finetune threshold module 150B may determine an index j, which is the maximal index corresponding to a precision value p_calc>=p_req. Finetune threshold module 150B may denote the precision of the corresponding prefix by p(j).
(111) In a third phase, denoted herein as “phase 3”, finetune threshold module **150**B may perform a process denoted herein as “tail analysis”, configured to determine that a drop in precision below index j is statistically likely to represent a trend rather than a local drop or ‘glitch’, as elaborated below. Finetune threshold module **150**B may repeat step 3a through 3d, until the required threshold **151**B is found at step 3d:
(112) Phase 3
(113) {
(114) 3a. If the tail size, defined as a difference between the number of samples (e.g., n_samp, in this example 300) and the predetefined minimal number of samples (e.g., MS, in this example 250) is less than a predefine number (e.g., 20, in this example 300−250=50>20), then finetune threshold

module 150B may go back to step 1a (e.g., sample s utterances from the next "unused" bin, bi, and increment i (i=i+1).
3b. Let k be the number of true predictions in the tail (e.g., the last 300−250=50 samples). Finetune threshold module 150B may compute the p-value, for at most k successes out of n_samp−j trials, assuming binomial distribution with success probability p_req. This p-value may represent the null hypothesis that the precision in the tail area is p_req or higher, and is computed using the binomial cumulative distribution function (CDF).
3c. If the p-value is greater than 0.05 (accept the null hypothesis) then finetune threshold module 150B may draw s samples from a new bin b.sub.i and, and finetune threshold module 150B may go back to step 2a. The motivation for this step may be that since the null hypothesis was accepted (e.g., a drop in the calculated precision p_calc was detected as a false alarm) finetune threshold module 150B may draw a minimal, additional number of samples from a new bin, and realculate the point where p_calc goes below p_req. 3d. If the p-value is <=0.05 (reject the null hypothesis) then finetune threshold module 150B may return prediction threshold value 151B as the score of the sample at index j.
56
(115) It may be appreciated that upon termination, the C=95% confidence interval is [p_calc—e,p_calc+e], where: e (e.g., the margin of error) may be calculated according to equation Eq. 4, below:
e=Z0.95×√{square root over (*p*_calc(1−*p*_calc)/*n*_samp)} Eq. 4 where Z0.95 is the z-score at 0.95.
(116) Taking as an example a required level of precision p_req=0.8, and since p_calc is expected to be very close to p_req, Eq. 4 may produce:
e˜1.96×√{square root over (0.8×0.2/250)}=0.049, which means that given a target precision of 0.8, and sample size of 300, the probability of the actual precision of the entire population to be contained within the range [0.751,0.849] is at least 0.95.
(117) Reference is now made to FIG. **4** is a flow diagram, depicting a method of finetuning automated sentiment classification by at least one processor (e.g., processor **2** of FIG. **1**), according to some embodiments of the invention.
(118) As shown in step S**1005**, the at least one processor may receive a first ML model M.sub.0 (e.g., ML model **120**A of FIG. **2**). First ML model M.sub.0 may be initially trained, or pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset **23** and/or validation set **25**.
(119) As shown in step S**1010**, the at least one processor may associate one or more instances of model M.sub.0 with, or deploy one or more instances of model M.sub.0, at one or more corresponding sites. These one or more sites may include, for example, a website, a call center, an online chat room, and the like, that may belong to, or be associated with one or more corresponding organizations, clients and/or companies. It may be appreciated that pretraining the first ML model M.sub.0 according to annotated training dataset **23** and/or validation set **25** may be performed at a site of a vendor of ML model M.sub.0, and may be done before deployment or association of ML model M.sub.0 to the one or more sites.
(120) As shown in step S**1015**, for at least one (e.g., each) ML model M.sub.0 instance, the at least one processor may receive at least one utterance **143** via the corresponding site. For example, the at least one utterance may be a vocal or verbal utterance, received from a customer via a phone call with a site that is a call center. In another example, the at least one utterance may be a textual utterance, received from a customer via site that is an online chat, or an organizational web page, and the like.
(121) As shown in step S**1020**, for at least one (e.g., each) ML model M.sub.0 instance, the at least one processor may obtain at least one data element of annotated feedback **145**, corresponding to the at least one utterance. For example, the at least one data element of annotated feedback **145** may

include a labeling or annotation relating to sentimental classification of the at least one utterance **143**. For example, annotated feedback **145** may include supervisory feedback regarding classification of at least one utterance **143** by ML model **120**A, such as a True Positive (TP) annotation, a True Negative (TN) annotation, a False Positive (FP) annotation and a False Negative (FN) annotation. Additionally, or alternatively, at least one data element of annotated feedback **145** may include labeled or annotated data such as "ground-truth" sentiment classification (e.g., "positive sentiment", "negative sentiment") of the at least one utterance **143**.
(122) As shown in step S**1025**, for at least one (e.g., each) ML model M.sub.0 instance **120**A, the at least one processor may retrain, or fine-tune training of ML model M.sub.0, to produce a second ML model M.sub.1 (e.g., ML model **120**B of FIG. **2**). Retraining of ML model **120**A may be based on a second annotated training dataset, wherein the second annotated training dataset may include the first annotated training dataset **23** and the at least one annotated feedback data element **145**.
(123) As shown in step S**1030**, for at least one (e.g., each) site (e.g., and/or corresponding ML model M.sub.0 instance **120**A), the at least one processor may use the second ML model M.sub.1 **120**B to classify utterances, e.g., utterances **143** received at the associated site, according to one or more sentiment classes (e.g., according to a "positive sentiment" class, a "neutral sentiment" class, and a "negative sentiment" class).
(124) Reference is now made to FIG. **5** which is a flow diagram, depicting another method of finetuning automated sentiment classification by at least one processor (e.g., processor **2** of FIG. **1**), according to some embodiments of the invention.
(125) As shown in step S**2005**, the at least one processor may receive a first ML model M.sub.0 (e.g., ML model **120**A of FIG. **2**). First ML model M.sub.0 may be initially trained, or pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset **23** and/or validation set **25**. Additionally, or alternatively, ML model M.sub.0 may be associated with at least one prediction threshold (e.g., initial threshold **151**A of FIG. **2**).
(126) As shown in step S**2010**, the at least one processor may associate one or more instances of model M.sub.0 with, or deploy one or more instances of model M.sub.0 at one or more corresponding sites. These one or more sites may include, for example, a website, a call center, an online chat room, and the like, that may belong to, or be associated with one or more corresponding organizations, clients and/or companies, as elaborated herein.
(127) As shown in step S**2015**, for at least one (e.g., each) ML model M.sub.0 instance **120**A, the at least one processor may receive at least one utterance via the corresponding site, as elaborated herein (e.g., in relation to FIG. **4**).
(128) As shown in step S**2020**, for at least one (e.g., each) ML model M.sub.0 instance, the at least one processor may obtain at least one data element of annotated feedback **145**, corresponding to the at least one utterance, as elaborated herein (e.g., in relation to FIG. **4**).
(129) As shown in step S**2025**, for at least one (e.g., each) ML model M.sub.0 instance, the at least one processor may automatically calibrate the at least one prediction threshold **151**B according to a required level of precision (e.g., denoted herein as p_req), and based on the at least one annotated feedback data element **145**, as elaborated herein (e.g., in relation to FIG. **3**).
(130) As shown in step S**2030**, for at least one (e.g., each) site, the at least one processor may use the at least one prediction threshold to classify utterances **143** as belonging to one or more sentiment classes, as elaborated herein.
(131) Embodiments of the invention may include a practical application for performing ML-based natural language processing.
(132) Embodiments of the invention may include several improvements over currently available technology of ML-based classification, and especially ML-based natural language processing, and sentiment analysis.
(133) For example, embodiments of the invention may improve ML-based sentiment analysis technology by refining or retraining a pretrained sentiment analysis ML model, according to

specific environment, jargon and/or field of a specific site or organization at which the ML model is deployed. This retraining may be based on, or consider: (a) an initial training dataset upon which the ML model was initially pretrained, and (b) supervisory feedback data, pertaining to the specific site at which the ML model is deployed.
(134) In another example, embodiments of the invention may improve ML-based sentiment analysis technology by receiving a required level of precision (denoted herein as p_req), intelligently selecting a minimal, and yet sufficient number of utterances as samples for receiving human feedback, based on p_req and automatically fine-tuning a prediction threshold of a prediction layer of the ML model, based on this minimal number of samples.
(135) Unless explicitly stated, the method embodiments described herein are not constrained to a particular order or sequence. Furthermore, all formulas described herein are intended as examples only and other or different formulas may be used. Additionally, some of the described method embodiments or elements thereof may occur or be performed at the same point in time.
(136) While certain features of the invention have been illustrated and described herein, many modifications, substitutions, changes, and equivalents may occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.
(137) Various embodiments have been presented. Each of these embodiments may of course include features from other embodiments presented, and embodiments not specifically described may include various features described herein.

Claims

1. A method of finetuning automated sentiment classification by a computing system to handle new vocabularies introduced by domains of new deployments, the method comprising: receiving a base machine learning (ML) model M.sub.0, pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset; deploying a respective instance of the base ML model M.sub.0 to each of one or more respective instantiation environments; for each deployed ML model M.sub.0 instance: receiving at least one respective utterance via a contact center application of the respective instantiation environment; obtaining at least one respective data element of annotated feedback, wherein the at least one respective data element of annotated feedback corresponds to the at least one respective utterance and to a respective prediction, determined by the deployed ML model M.sub.0 instance, of a sentiment class of the at least one respective utterance and includes a user determined classification of the at least one respective utterance and a corresponding feedback type; retraining the deployed ML model M.sub.0 instance to produce a second respective ML model M.sub.1 based on a second respective annotated training dataset that comprises the first annotated training dataset and the at least one respective data element of annotated feedback, wherein the at least one respective data element of annotated feedback is weighted between five and ten times that of the first annotated training dataset when retraining the deployed ML model M.sub.0 instance to produce the second respective ML model M.sub.1; and using the second respective ML model M.sub.1, to classify utterances according to one or more sentiment classes.
2. The method of claim 1, wherein obtaining the at least one respective data element of annotated feedback is done interactively, via a user interface (UI) associated with the respective instantiation environment.
3. The method of claim 1, wherein the user determined classification of the at least one respective utterance comprises classification of the at least one respective utterance according to two or more sentiment classes.
4. The method of claim 1, further comprising: receiving an annotated validation dataset, comprising a plurality of utterances, and computing a corresponding plurality of confidence scores, wherein

each confidence score represents a certainty level of classification of an utterance to a sentiment class; sorting the utterances of the validation dataset in an ordered list, according to the computed confidence scores; for one or more entries of the ordered list, computing a value of classification precision; identifying an index of the ordered list that corresponds to a required classification precision value; and determining the confidence score of the identified index as a prediction threshold for classifying utterances as belonging to the sentiment class.
5. A method of finetuning automated sentiment classification by a computing system to handle new vocabularies introduced by domains of new deployments, the method comprising: receiving a base ML model M.sub.0, pretrained to perform automated sentiment classification of utterances, wherein the base model M.sub.0 is associated with at least one prediction threshold; deploying a respective instance of the base model M.sub.0 to each of one or more respective instantiation environments; for each deployed ML model M.sub.0 instance: receiving at least one respective utterance via a contact center application of the respective instantiation environment; obtaining at least one respective data element of annotated feedback, wherein the at least one respective data element of annotated feedback corresponds to the at least one respective utterance and to a respective prediction, determined by the deployed ML model M.sub.0, of a sentiment class of the at least one respective utterance and includes a user determined classification of the at least one respective utterance and a corresponding feedback type; automatically calibrating the at least one prediction threshold according to a required level of precision for the respective instantiation environment to produce at least one calibrated prediction threshold, and based on the at least one respective data element of annotated feedback; and using the at least one calibrated prediction threshold to classify utterances as belonging to one or more sentiment classes.
6. The method of claim 5 wherein obtaining the at least one respective data element of annotated feedback comprises: presenting one or more utterances to a user, via a UI; and obtaining the at least one respective data element of annotated feedback from the user via the UI, wherein the annotated feedback comprises classification of the presented one or more utterances according to two or more sentiment classes.
7. The method of claim 6, wherein presenting one or more utterances comprises: calculating a plurality of confidence scores to a corresponding plurality of utterances, according to each utterance's pertinence to a sentiment class; selecting a subset of utterances of the plurality of utterances for feedback based on their respective confidence scores; and presenting the selected subset of utterances via the UI.
8. The method of claim 7, wherein selecting one or more utterances for feedback comprises, for each sentiment class: selecting a first subset of utterances that surpass a selection threshold from the plurality of utterances; and randomly selecting a second subset of utterances from the first subset; and presenting the second subset of utterances via the UI.
9. The method of claim 8, wherein the prediction threshold is specific for each sentiment class.
10. The method of claim 8, wherein the size of the second subset of utterances is limited by a predefined subset size, that is specific for each sentiment class.
11. The method of claim 8, wherein the second subset of utterances comprises 300 utterances or less.
12. The method of claim 8, wherein calibrating the at least one prediction threshold comprises: calculating one or more confidence scores, corresponding to the one or more utterances of the second subset, according to each utterance's pertinence to a sentiment class; sorting the plurality of utterances of the second subset in a descending ordered list, according to the confidence scores; for one or more entries of the ordered list, computing a value of classification precision based on the annotated feedback; identifying an index of the ordered list that corresponds to a required classification precision value; and determining the confidence score of the identified index as the value of the prediction threshold.
13. The method of claim 8, wherein the predefined subset size is selected so as to guarantee that the

prediction threshold corresponds to ground truth precision having a confidence value of at least 95% within a margin of error of 0.05.
14. The method of claim 5, wherein the sentiment classes are selected from a list consisting of positive sentiments and negative sentiments.
15. A system for finetuning automated sentiment analysis to handle new vocabularies introduced by domains of new deployments, wherein said system is associated with a specific site and comprising: (a) a non-transitory memory device storing instruction code, and (b) a processor associated with the memory device, wherein the processor is configured to execute the instruction code so as to: receive a base ML model M.sub.0, pretrained to perform automated sentiment classification of utterances, based on a first annotated training dataset; deploy an instance of the base ML model M.sub.0 to an instantiation environment associated with the specific site; receive at least one utterance by the instantiation environment via the site; obtain at least one data element of annotated feedback, wherein the at least one data element of annotated feedback corresponds to the at least one utterance and to a prediction, determined by the deployed ML model M.sub.0 instance, of a sentiment class of the at least one utterance and includes a user determined classification of the utterance and a corresponding feedback type; retrain the deployed ML model M.sub.0 instance to produce a second ML model M.sub.1 based on a second annotated training dataset, wherein the second annotated training dataset comprises the first annotated training dataset and the at least one data element of annotated feedback, wherein the at least one data element of annotated feedback is weighted between five and ten times that of the first annotated training dataset when retraining the deployed ML model M.sub.0 instance to produce the second ML model M.sub.1; and use the second ML model M.sub.1 to classify utterances according to one or more sentiment classes.
16. The method of claim 1, wherein obtaining at least one respective data element of annotated feedback that includes a corresponding feedback type comprises obtaining at least one respective data element of annotated feedback that includes a corresponding feedback type indicative of a true positive, a true negative, a false positive, or a false negative.